

COURSEFLIX

An Arabic-First Online Learning Platform

Full-stack LMS · RAG AI Tutor · Learning Interventions · Paymob Payments · Sales Analytics

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Introduction

CourseFlix is an Arabic-first, RTL online learning platform that connects teachers and students in a complete learning loop — from publishing course content to assessing, intervening, and paying for courses.

Student Experience

Enrolled-course dashboard, lessons with resume + attendance, server-graded quizzes, course-scoped AI Tutor, notifications, interventions, and checkout.

Teacher Experience

Course/section/lesson management, PDF upload with ingestion, quizzes, announcements, discussions, sales metrics, a limited Analytics Agent, and agent logs.

Assistant & Admin

Assistants share the teacher course/student surface; admins oversee users, courses, orders, quizzes, documents, notifications, and agent logs.

OVERVIEW

Platform Overview

A single platform covering the whole learning loop — from how a course is structured to how students learn, get assessed, and receive help.

Publish

Teachers structure courses into sections and lessons, upload PDFs, and post announcements.

Practice

Students watch lessons with resume + attendance and take objective quizzes.

Assess

Server-side grading, interventions, and weak-concept reports flag struggles early.

Feedback

AI Tutor answers with citations, sales/analytics surface outcomes, and notifications keep everyone in sync.

User Journey

A full learning loop told from the student and teacher perspectives.

Student

Enroll — browse published courses and sign up

Learn — watch lessons, resume progress, gain attendance

Assess — take server-graded quizzes

Ask — query the AI Tutor with cited answers

Buy — check out with Paymob and get a receipt

Get help — interventions and support tickets

Teacher

Structure — courses → sections → lessons

Upload — PDFs become searchable by the Tutor

Assess — quizzes from a shared question bank

Monitor — progress reports and interventions

Sell — sales metrics and the Analytics Agent

Engage — announcements, discussions, support

OBJECTIVES

Project Objectives

Deliver an Arabic-First RTL learning experience based on the approved UI design.

Let students watch lessons, track progress, and get server-graded quiz results.

Add deterministic learning interventions that never auto-suspend a student.

Give teachers backend-sourced sales metrics and a limited Analytics Agent.

Let teachers structure courses into sections and lessons, upload PDFs, and post announcements.

Provide a grounded, honest AI Tutor that answers only from the teacher's documents with citations.

Enable secure course payments via Paymob with backend-sourced orders and receipts.

Keep everything traceable and privacy-safe (correlation IDs, redacted logs, PII-denylisted analytics).

Technologies Used

Frontend

React + TypeScript\nVite · React Router · TanStack Query\nTailwindCSS + custom design tokens\nArabic RTL responsive UI

Backend

NestJS + TypeScript\nREST API · DTO validation · guards\nTypeORM migrations (PostgreSQL)\nArgon2id + signed session cookies

AI & Data

ChromaDB vector store\nBullMQ + Redis job queue\nOpenAI embeddings/LLM (provider-agnostic)\nDeterministic mock providers for tests

Infra & Tests

npm workspaces monorepo\nDocker Compose (Postgres/Redis/Chroma)\nJest + Supertest · Vitest + RTL + MSW\ndev.sh one-command launcher

SYSTEM FEATURES

System Features

✓ **Auth & Roles**

Password + Google OTP sign-in, role guards, sessions

✓ **AI Tutor (RAG)**

Cited answers from course docs, honest no-answer

✓ **Notifications**

Filterable feed, mark-read, deep links

✓ **Sales & Analytics**

Backend metrics + limited Analytics Agent

✓ **Course Delivery**

Sections → lessons, resume playback, attendance

✓ **Video Q&A**

Transcript-grounded assistant per video

✓ **Community**

Discussions, announcements with attachments

✓ **Support**

Tickets routed to support staff

✓ **Graded Quizzes**

Server-side grading, question bank, one attempt

✓ **Interventions**

Weak-concept reports + mini-quiz, no auto-suspend

✓ **Paymob Payments**

Real checkout with Paymob gateway, backend order truth

✓ **Admin Oversight**

Users, courses, orders, logs, notifications

ARCHITECTURE

System Architecture

Web (React)

Vite SPA · RTL UI

Worker (NestJS)

BullMQ ingestion

API (NestJS)

REST · Guards · DTO validation

PostgreSQL

TypeORM · migrations

Redis

BullMQ job queue

ChromaDB

vector store · isolation

AI Providers

OpenAI LLM + embeddings

Backend Overview — Auth, Courses, Lessons

Endpoint	Method	Path	Description
/api/auth	POST	/login	Sign in with email/password, sets session cookie
/api/auth	POST	/register	Student self-registration + verification OTP
/api/auth/oauth	GET	/google	Start Google "Continue with Google" flow
/api/auth/otp	POST	/verify	Redeem OTP (register / google_oauth), open session
/api/courses	GET	/:courseId	Course detail (enrolled student / owning teacher)
/api/courses	POST	/:courseId/sections	Create a section in a course
/api/lessons	PATCH	/:lessonId/progress	Report video progress (heartbeat, attendance)
/api/me	GET	/	Current authenticated user

Backend Overview — Quizzes, Tutor, Commerce

Endpoint	Method	Path	Description
/api/quizzes	GET	/:quizId	Quiz detail (answer key never serialized)
/api/quizzes	POST	/:quizId/submissions	Submit answers, server-side grading (409 on retry)
/api/courses	POST	/:courseId/tutor/messages	Ask the AI Tutor (RAG citations or no-answer)
/api/checkout	POST	/orders	Create draft order with server-set price
/api/checkout	POST	/orders/:id/confirm	Confirm order, start Paymob payment flow
/api/orders	GET	/:orderId	Fetch order + receipt (backend financial truth)
/api/teacher/sales	GET	/summary	Revenue, order count, best-seller (paid only)
/api/teacher/analytics	POST	/questions	Analytics Agent (3 allowlisted intents)

Frontend — Learning Flow

Login / Register

Email+password or Google OTP step; role-guarded redirect; reset password.

Lesson Player

Video playback with resume, throttled heartbeat, and attendance progress.

AI Tutor Chat

Ask about the course; see cited answers or an explicit no-answer.

Student Dashboard

Enrolled courses with progress and status; course detail with sections/lessons.

Quiz Page

Objective quiz, one attempt, immediate server-graded result.

Teacher Dashboard

Course/section/lesson management and progress overview.

Frontend — Community, Commerce, Admin

Files Tab

Upload PDFs, watch ingestion status (pending → processing → done/failed).

Interventions

Weak-concept report and mini-quiz views for students and teachers.

Sales & Analytics

Sales summary and the limited Analytics Agent UI.

Notifications

Filterable feed, mark one/all read, deep-link to the exact entity.

Checkout & Receipt

Deterministic payment flow with loading/declined/success states.

Admin Pages

Users, courses, orders, quizzes, documents, notifications log, agent logs.

Screens — Auth

Sign in with email/password or Google OTP; self-registration wizard.

Login

Register

Screens — Student Home

Enrolled courses with progress and a structured course detail page.

Student Dashboard

Course Detail

Screens — Learning

Video lessons with resume + attendance, and server-graded quizzes.

Lesson Player

Quiz

Screens — AI & Engagement

The course-scoped AI Tutor and the community / discussions surface.

AI Tutor Chat

Community

Screens — Account & Alerts

Notifications with deep links, and theme/account preferences.

Notifications

Settings

TEACHER WORKSPACE

Teacher Workspace

Course management, students, and backend-sourced sales metrics.

Teacher Dashboard

Sales & Analytics

FUTURE

Future Enhancements

- 1** Paymob expansion — recurring plans, coupons, and multi-currency support.
- 2** Advanced AI Tutor — conversation memory summaries and expanded grounding.
- 3** AI quiz generation — generate quizzes directly from uploaded documents.
- 4** Progress reports — automated weekly teacher reports for at-risk students.
- 5** Mobile app — React Native or Flutter clients.
- 6** Deployment on Render/AWS — Docker containers with managed Postgres and Redis.
- 7** Notifications delivery — email/SMS channels beyond in-app notifications.
- 8** Expanded analytics — richer charts and more allowlisted intents.

CONCLUSION

Conclusion

CourseFlix demonstrates a complete, production-hardened learning platform:

Grounded AI — a cited, honest Tutor that answers only from the teacher's documents.

Deterministic quality — server-side grading, versioned quizzes, and rule-based interventions that never auto-suspend.

Financial truth — backend orders as the single source of truth for sales and receipts.

Privacy-safe — correlation IDs, redacted logs, and PII-denylisted analytics.

It brings teachers and students into one coherent, measurable learning loop.

REFERENCES

References

GitHub Repository — github.com/Xmerlin7/CourseFlix

API Documentation — OpenAPI / Postman collection

Database Schema — schemaV2.sql (PostgreSQL, 35 tables)

Project Plans — Sprint 1/2/3 plans and Scrum/Jira backlog

UI Design — approved ui5 handoff

Thank You!

Feel free to ask any questions